

## **STAT 7500 - Statistical Programming**

### **SAS Mini-Project**

For the project, please submit by **Saturday October 4<sup>th</sup>, 2025** via Blackboard as either...

- a) (preferred method) A single document that includes your responses to the questions, including the output, as well as your complete SAS code attached as an appendix, or
- b) Your SAS editor, that includes comments that you are asked to make, as well as your output.

Please make sure your name is somewhere in the document. These projects are to be done independently. You may use any online guides or resources you want, but please do not consult other students in the class or other people, except for Dr. Fitzgerald. You may use Google or ChatGPT (or other LLM) for coding questions, but please do not ask them to answer the question itself. Please respond to the questions as briefly as possible, but with complete answers. I strongly recommend that you not delay on starting the assignment, especially reading in the data.

**Background:** We are interested in exploring the relationship between graduating college and income for Pennsylvanians. You have found two files to help you accomplish this task. The first is a list of median household income by zip code for the entire US (as well as population and mean household income). These data can be found in the file **Median Income by Zip Code in US.xlsx**. The second file lists the percent of residents of each zip code in Pennsylvania who are college graduates. The file also contains the city name associated with that zip code. These data can be found in the file **PA College Graduation By Zip Code.xlsx**.

The specific tasks to complete are on the following page...

Complete the following tasks:

- 1) Read in each file into SAS.
- 2) Merge the two files together by zip code. When doing so, create three output datasets – one called *match* that contains only those zip codes that were in both files, a second one called *noinc* that includes zip codes that don't have income data, and a third one called *nograd* that includes zip codes that don't have college graduation data.
- 3) How many zip codes are in each of the three datasets that you created above? For the *noinc* and *nograd* datasets, speculate as to why these values did not match. You might need to look at the zip codes themselves to figure this out. (I found a website called zip-codes.com that helped me answer this question. If you enter the zip code in the “Free Zip Code Finder” section, information on that zip code will appear. If you are not familiar with how US Zip codes work, please come see me and I'm happy to offer some guidance).
- 4) Using only the *match* dataset...
  - a) Calculate the quartiles of the college graduation rate variable.
  - b) Create a new variable called *CollGradGroup* which takes on the following values:
    - 4 if the graduation rate for that zip code is greater than the third quartile
    - 3 if the graduation rate for that zip code is between the median & the third quartile
    - 2 if the graduation rate for that zip code is between the first quartile & the median
    - 1 if the graduation rate for that zip code is lower than the first quartile
    - . (that's a missing value) if the graduation rate is not givenFormat the *CollGradGroup* variable to name the groups “high”, “med-high”, “med-low”, and “low”, respectively. Add a label to the variable as well.
  - c) You are interested in calculating the mean value of the median income variable as well as the mean value of the population variable separately for each *CollGradGroup* value. (Thus, you want the mean income for high, med-high, med-low, and low as well as the mean population for each of them as well). Create this in three ways:
    - i) As a SAS dataset, where the variables are *meaninc* and *meanpop*.
    - ii) In a table using PROC TABULATE.
    - iii) In a scatterplot, where  $x = \text{meanpop}$  and  $y = \text{meaninc}$ . Comment on the pattern that you see in the graph.